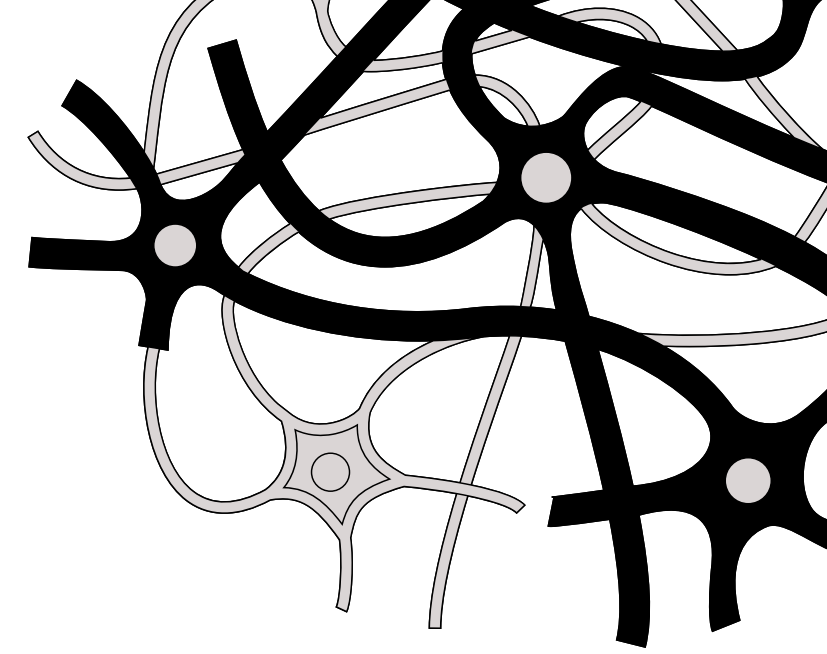
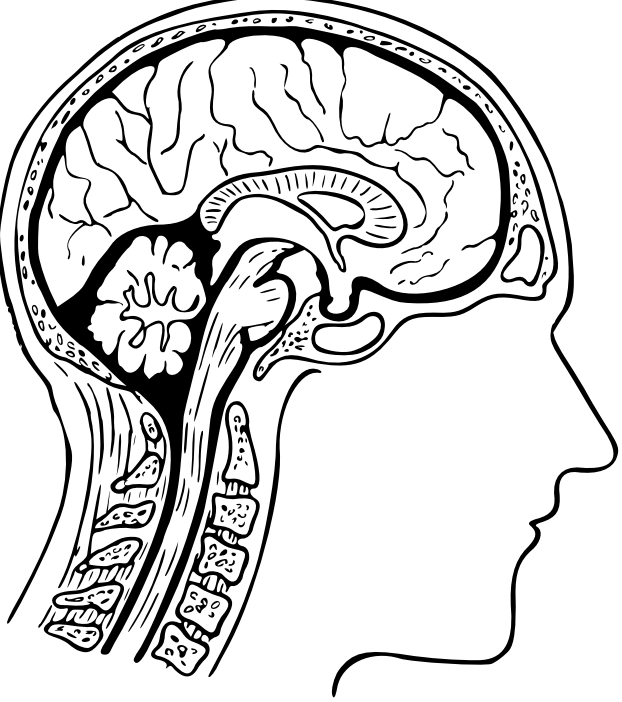
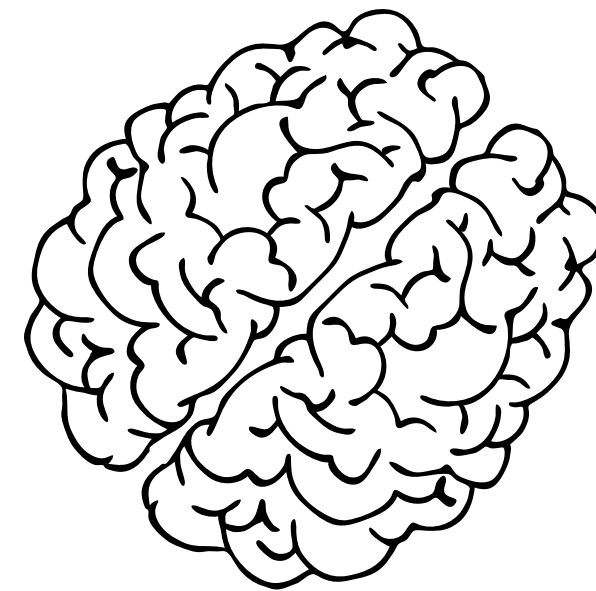
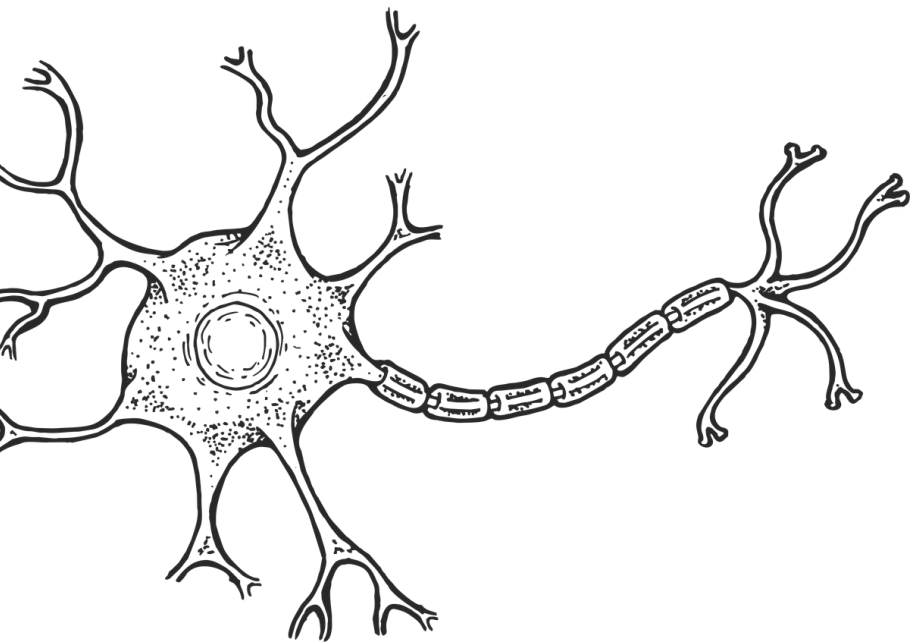




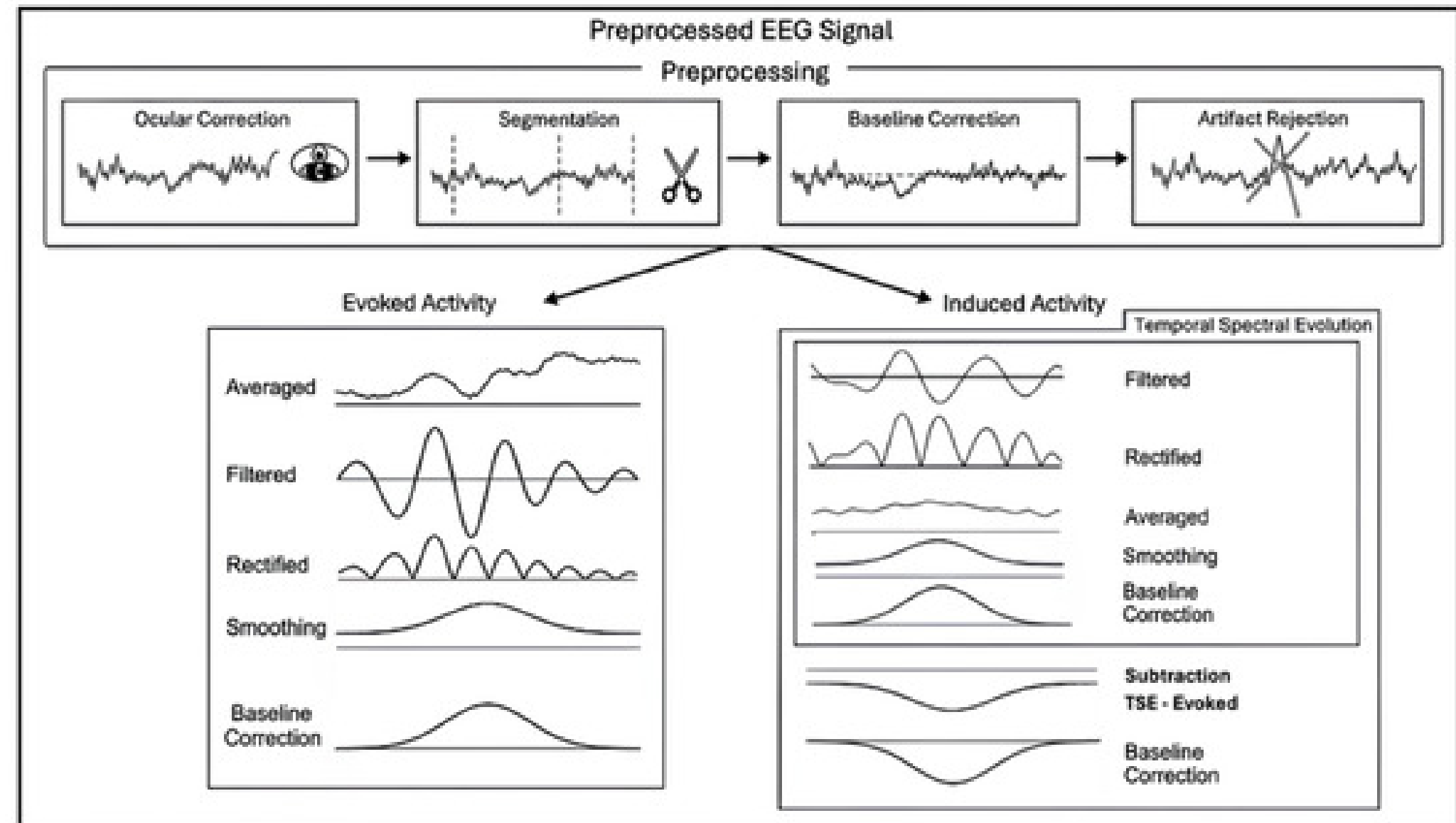
Session 7: Preprocessing II: Epoching and Independent Component Analysis (ICA)

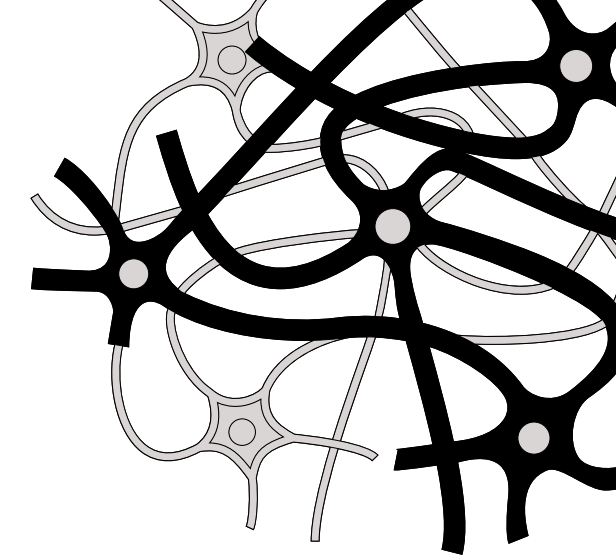
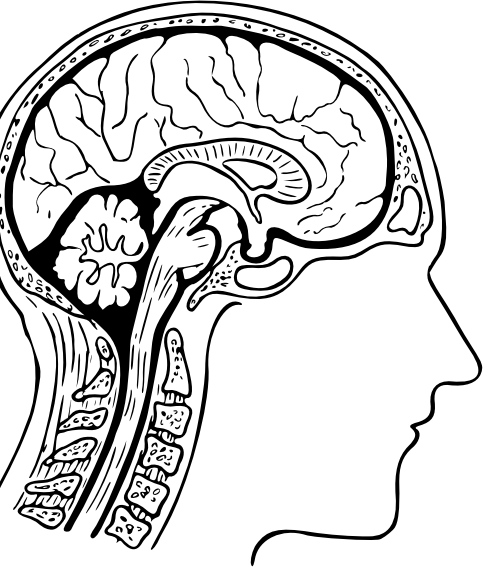
presented by Lynn



Preprocessing II

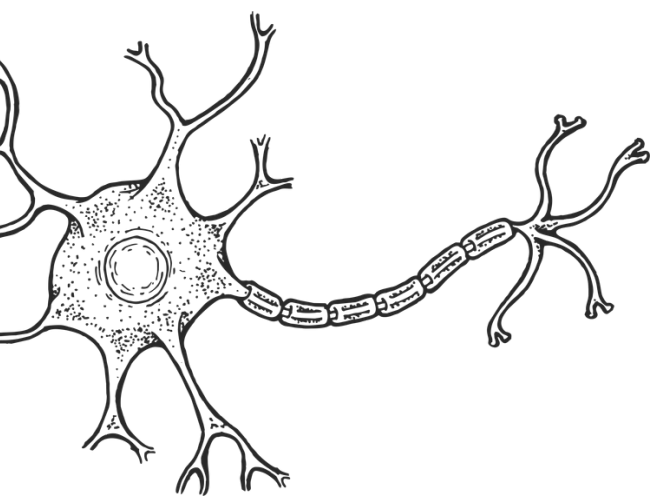
- From Last session:
 - We established a stable reference.
 - We applied notch filters to remove line noise (50/60 Hz).
 - We filtered out slow environmental drifts.
- Goal Today:
 - How do we isolate specific event responses?
 - How do we remove physiological artifacts (like blinks) that share the exact same frequency band as our neural signals without destroying the data?





Part A: Epoching

Slicing the continuous recording into discrete, manageable analytical windows.

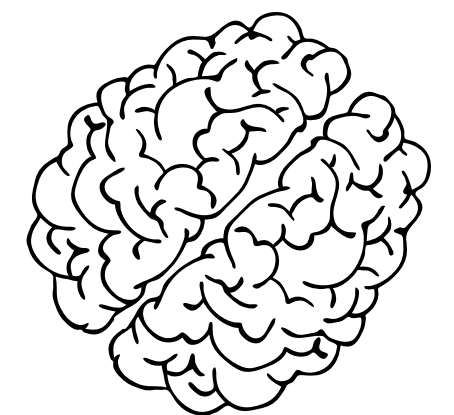


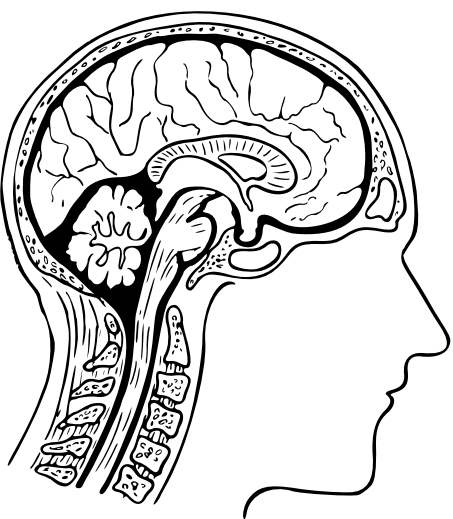
Part B: ICA Concept

Understanding the math and geometry behind blind source separation.

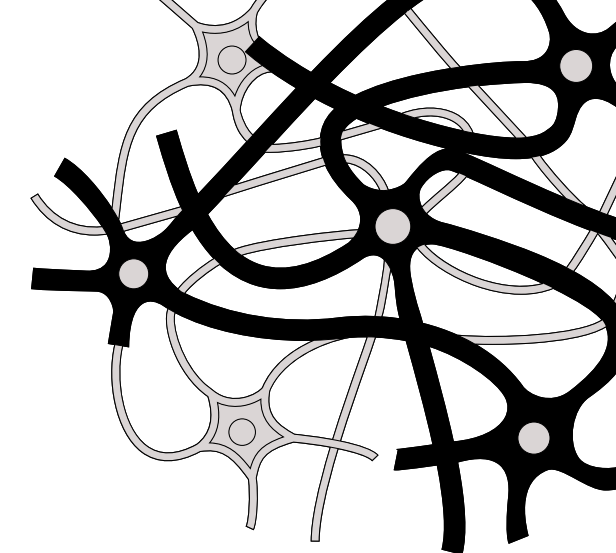
Part C: Live Coding

Applying ICA in MNE-Python to automatically detect and reject ocular artifacts.





Part A: Epoching

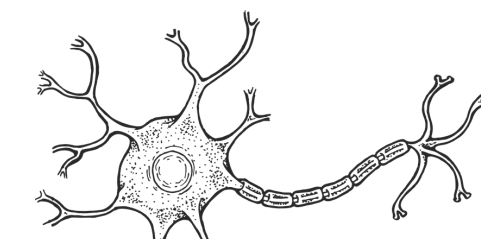
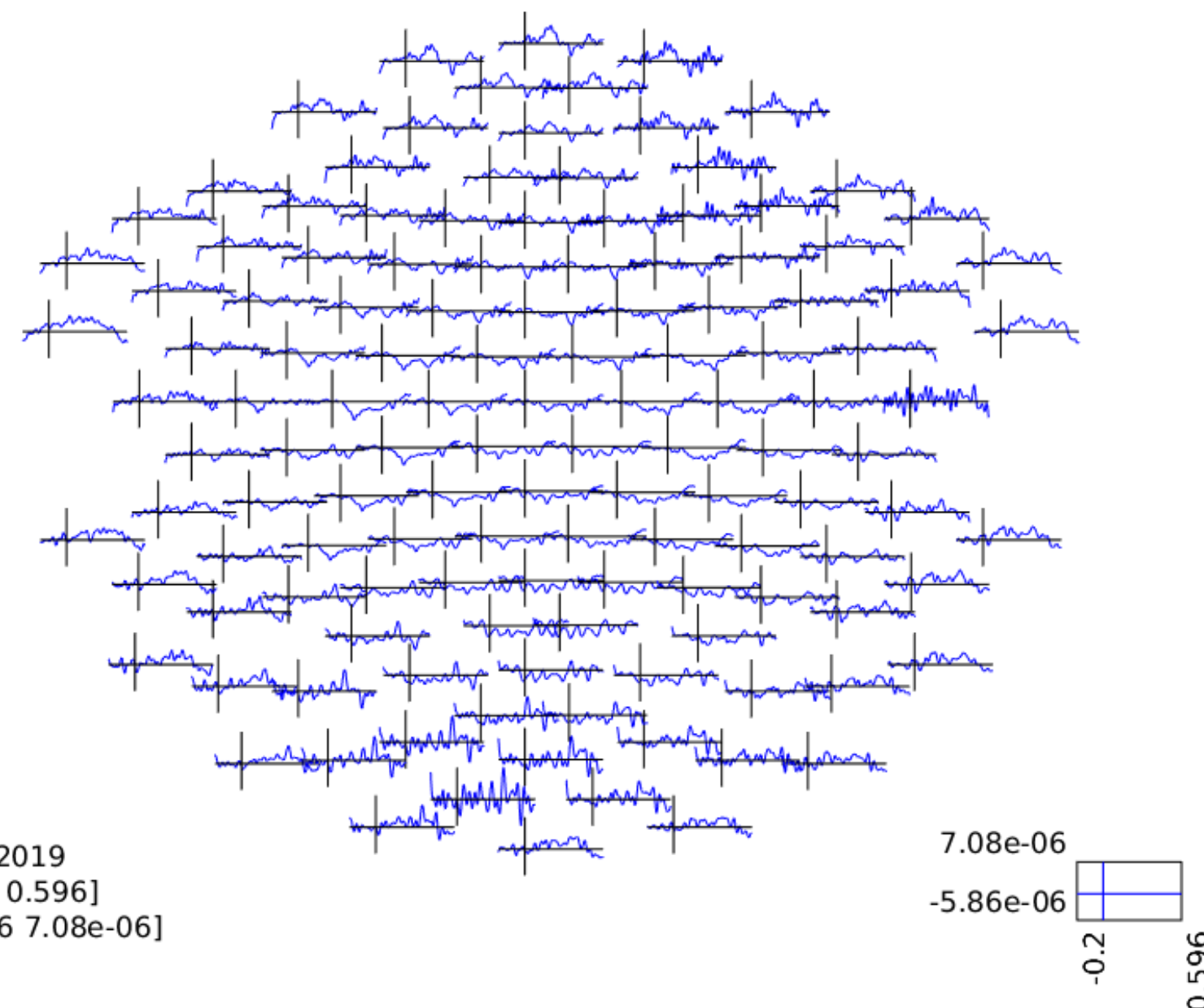


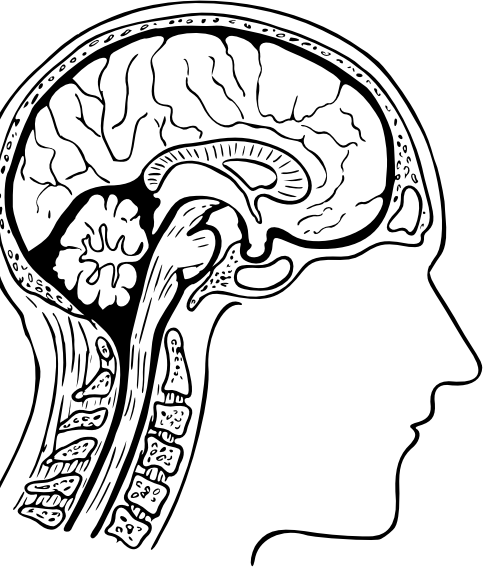
Problem with Continuity

- EEG is recorded continuously for minutes or hours, resulting in massive, unbroken matrices of voltages. However, cognitive experiments happen in discrete milliseconds.

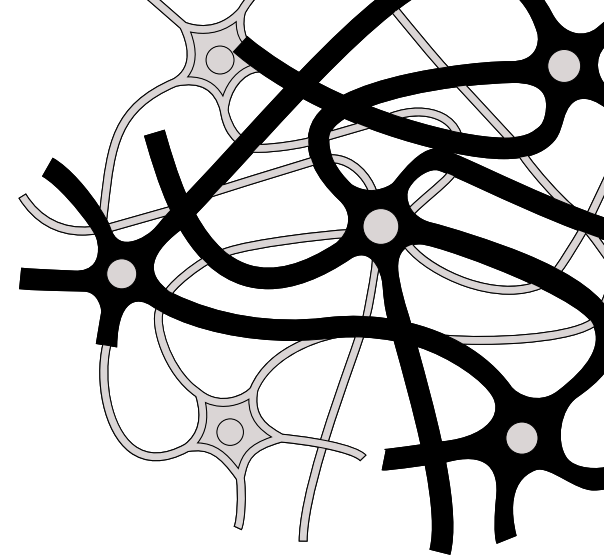
Event triggers & Slicing

- When a stimulus is shown, the experimental software sends a digital pulse to the amplifier ($t = 0$).
- Epoching cuts the continuous data into fixed, identical time windows tightly locked to these triggers, allowing us to compare apples to apples.



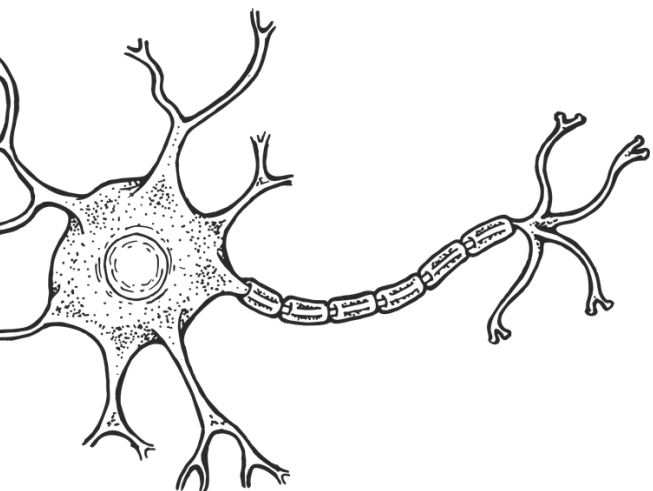


What defines an epoch



1. Time Window

- Defined strictly relative to trigger onset ($t = 0$).
- Example: We extract -200 ms prior to the stimulus, extending to +800 ms after.

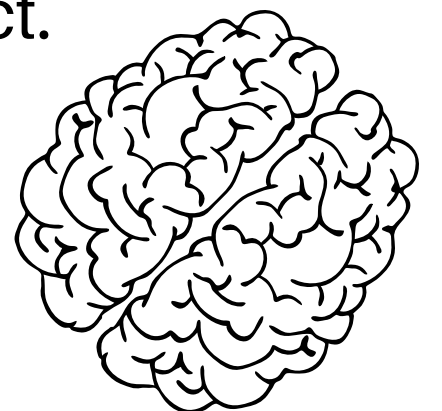


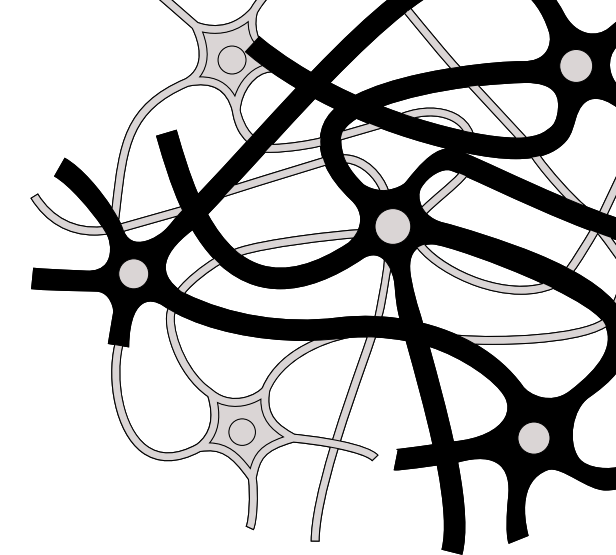
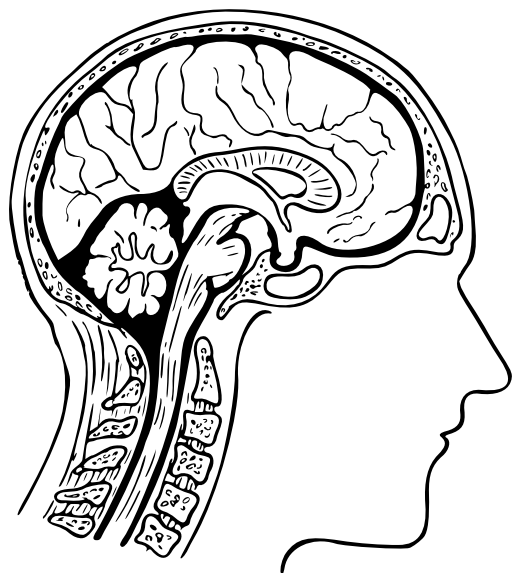
2. Baseline correction

- Subtracting the mean voltage of the pre-stimulus window (e.g., -200 to 0 ms) from the entire epoch.
- This zeroes out the signal at stimulus onset, correcting slow DC drifts.

3. Epoch Rejection

- Dropping specific epochs based on extreme peak-to-peak voltage swings.
- Example: If an epoch exceeds a $150 \mu\text{V}$ differential, it's discarded as a movement artifact.





Part B: The Mixing Problem

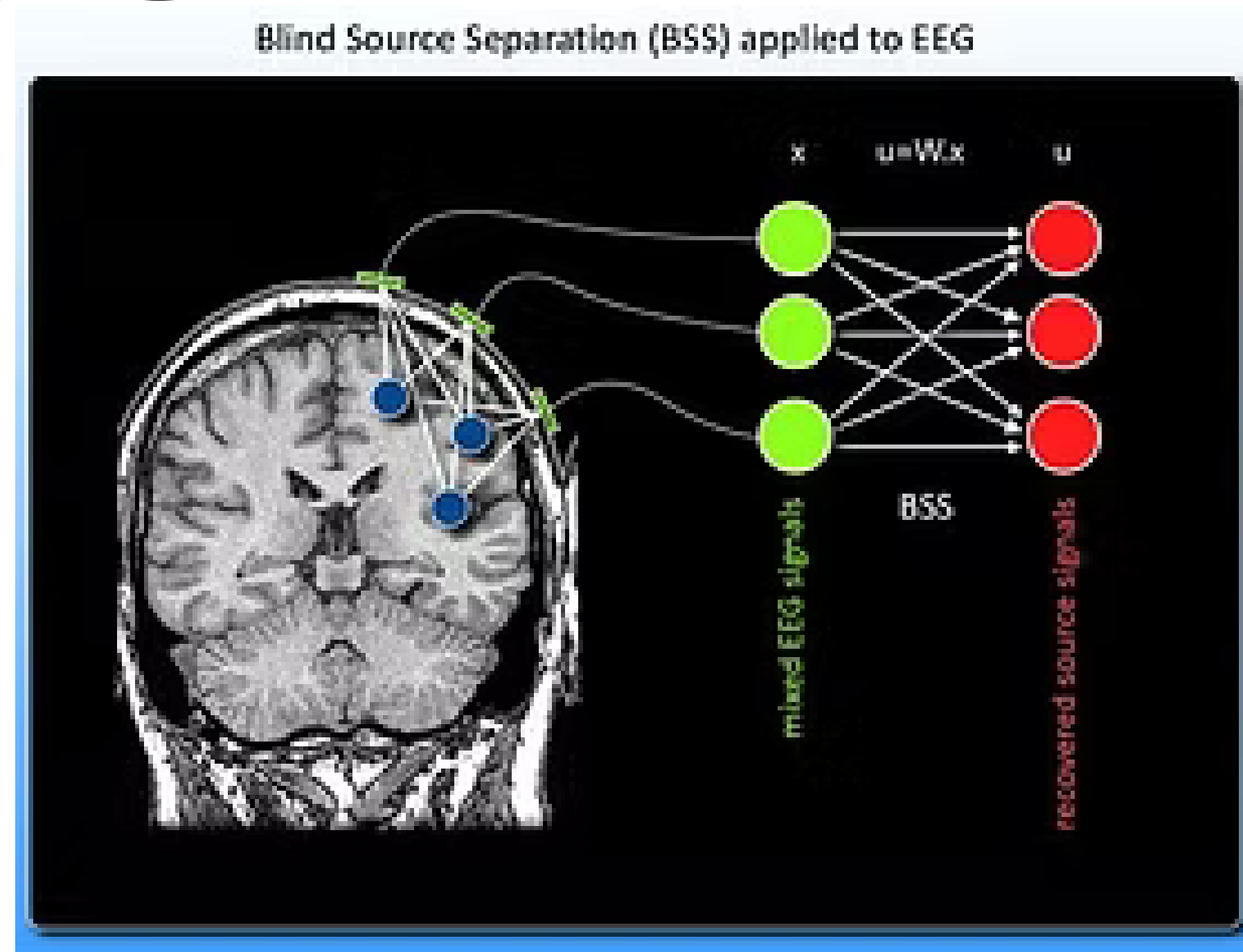
The Scalp is a Mixer

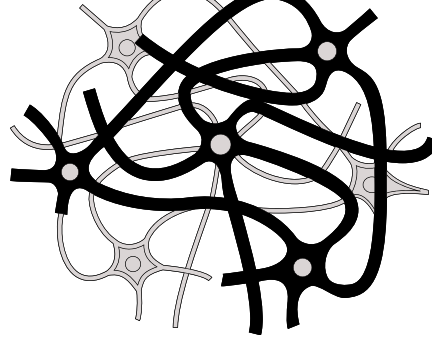
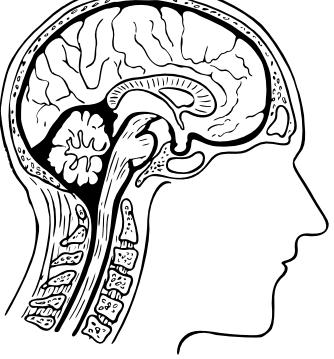
Every single electrode on the scalp picks up a linear, immediate mixture of overlapping electrical sources. It does not know where the signal came from.

- **Source 1:** Visual cortex alpha rhythm
- **Source 2:** Eye blink (Ocular dipole)
- **Source 3:** Heartbeat (Cardiac pulse)

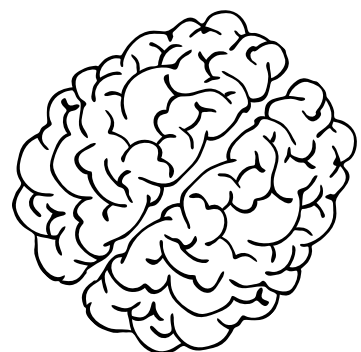
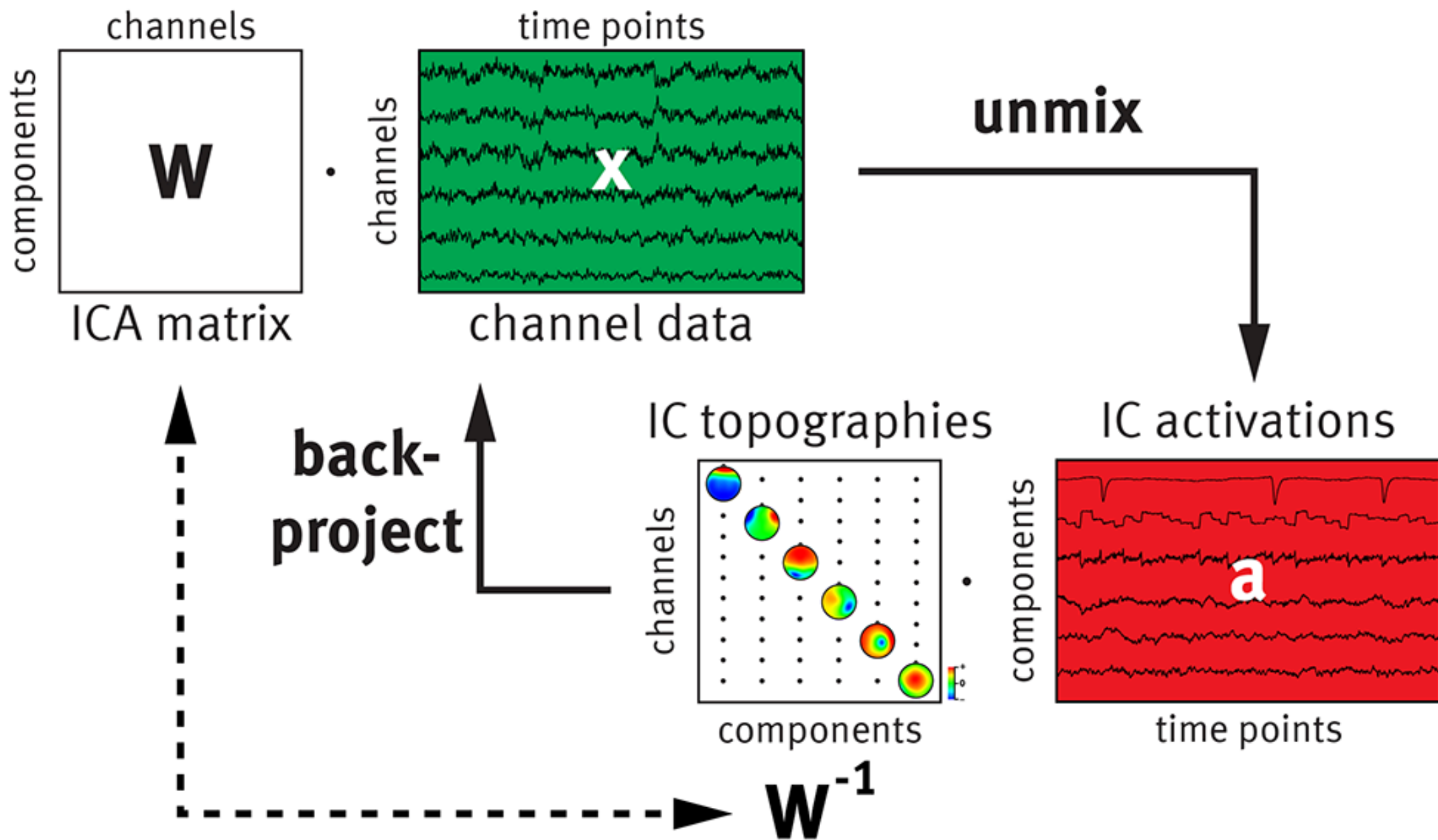
Why Bandpass Fails

- An eye blink generates massive power in the Delta/Theta bands. If you use a standard frequency filter to remove the blink, you permanently delete genuine low-frequency brain waves.





Independent Component Analysis (ICA)



The Generative Model

$$X = AS$$

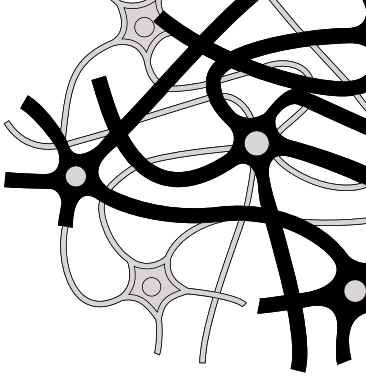
- X = Observed scalp data (Channels $\times$ Time)
- S = Independent latent sources (Components $\times$ Time)
- A = Mixing matrix (Scalp projection)

The Goal

- Compute the unmixing matrix W (which is A^{-1}) to recover the independent sources.

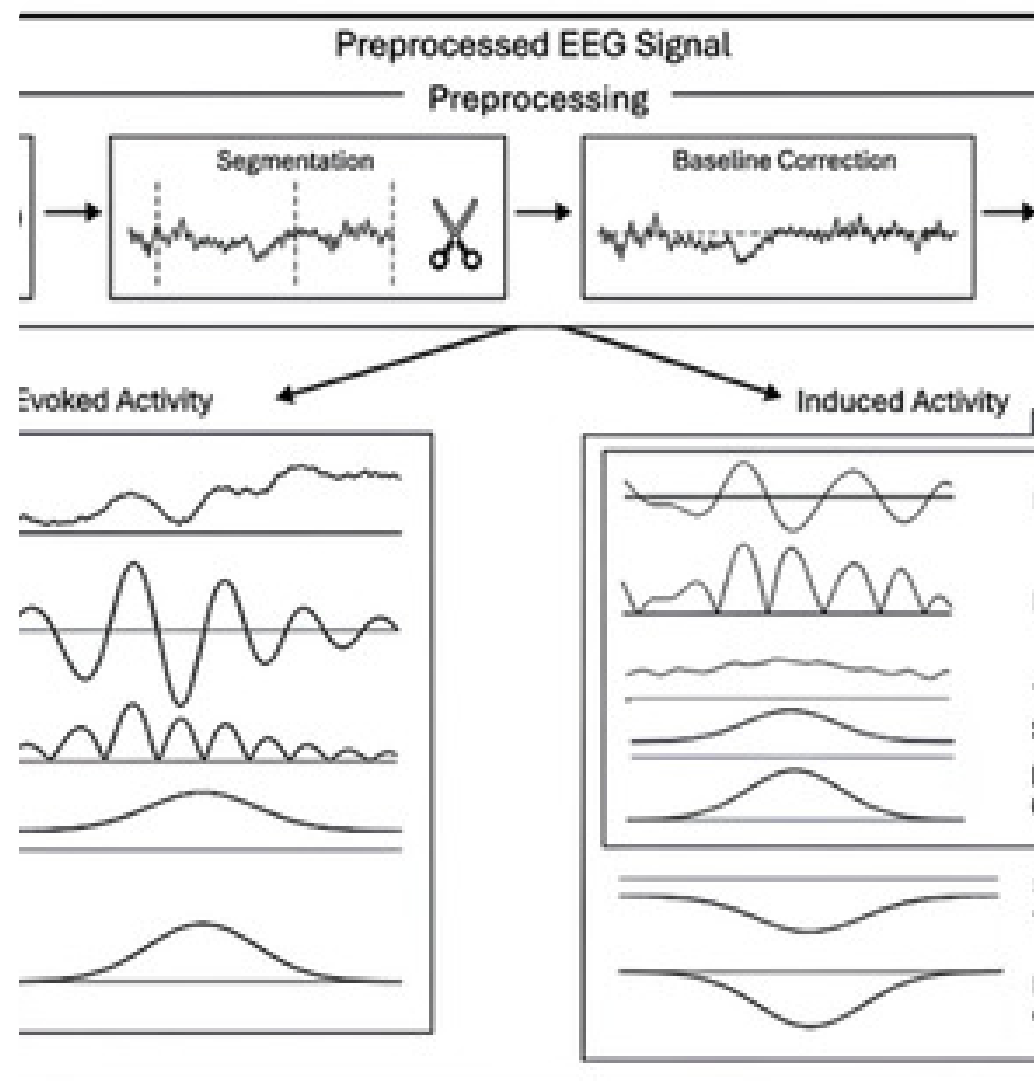
$$S = WX$$

Stationary Assumption

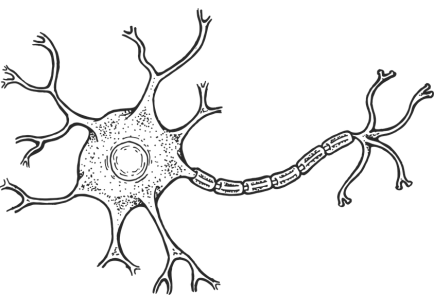


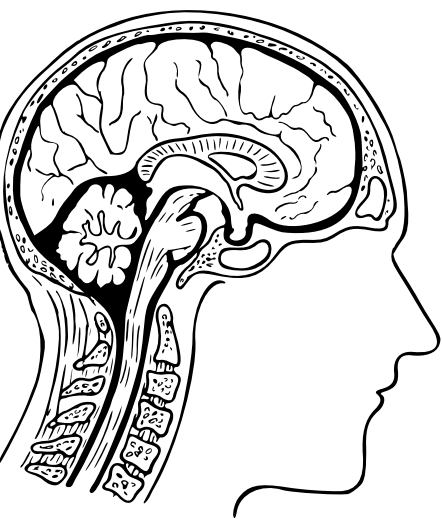
Why do we need a 1Hz Highpass Filter

ICA requires the statistical properties of the sources to remain constant over time (stationarity). Slow voltage drifts severely violate this mathematical assumption.

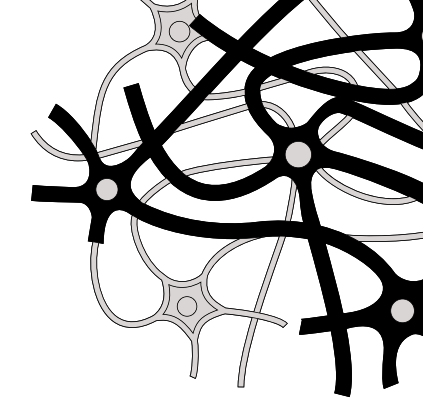


- Sweat and movement cause the "baseline" voltage to wander slowly.
- Without a 1 Hz filter, ICA components become "blurry" and fail to cleanly isolate distinct physical sources like blinks.
- **Crucial Step:** We filter a temporary copy of the data at 1 Hz just to compute the ICA weights, then we apply those clean weights back to the primary data.



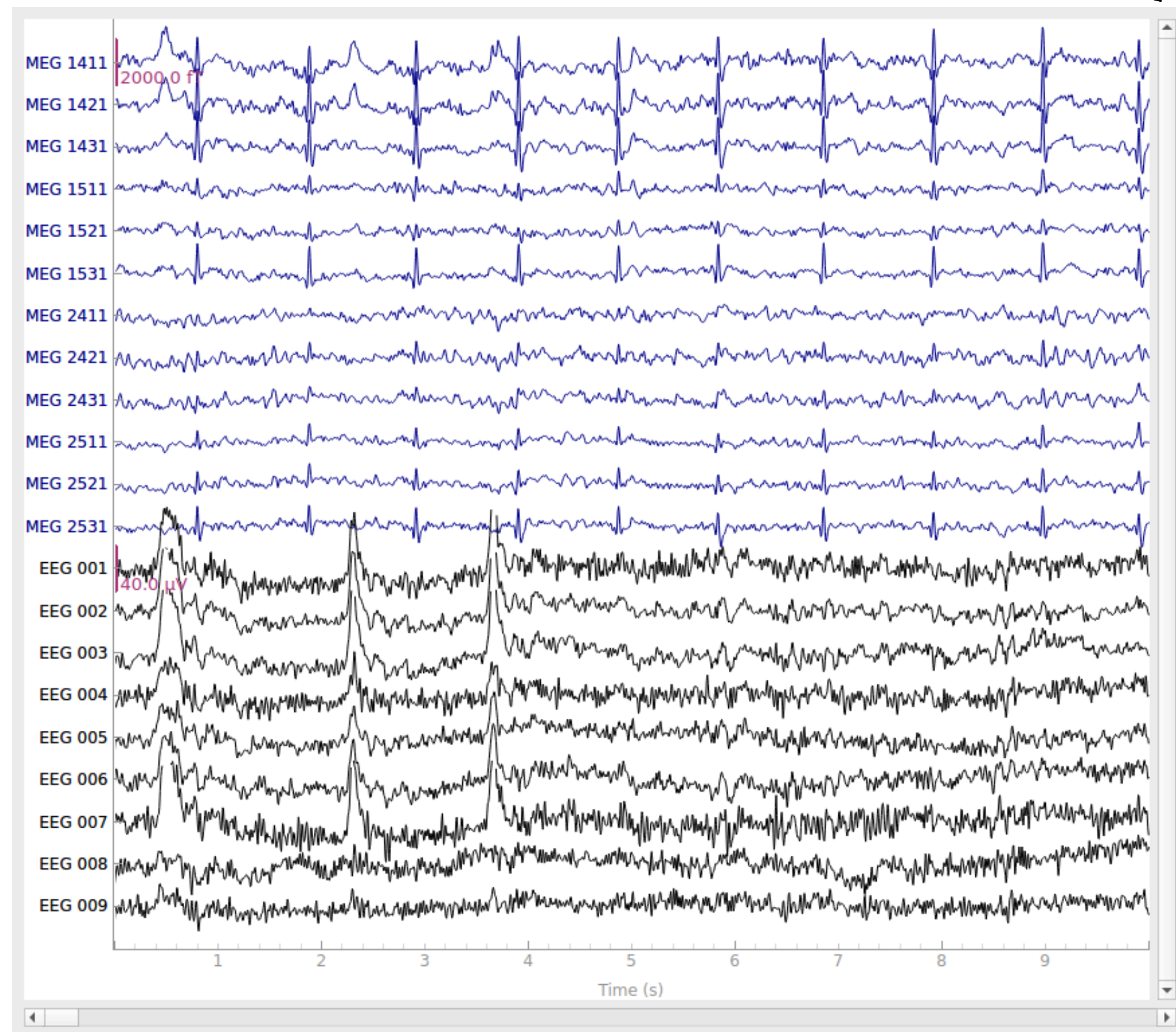


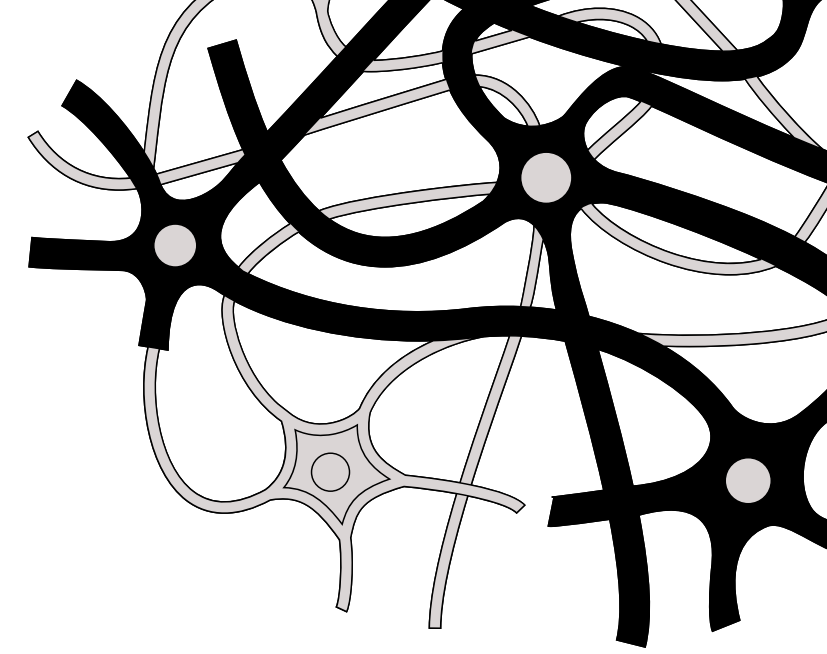
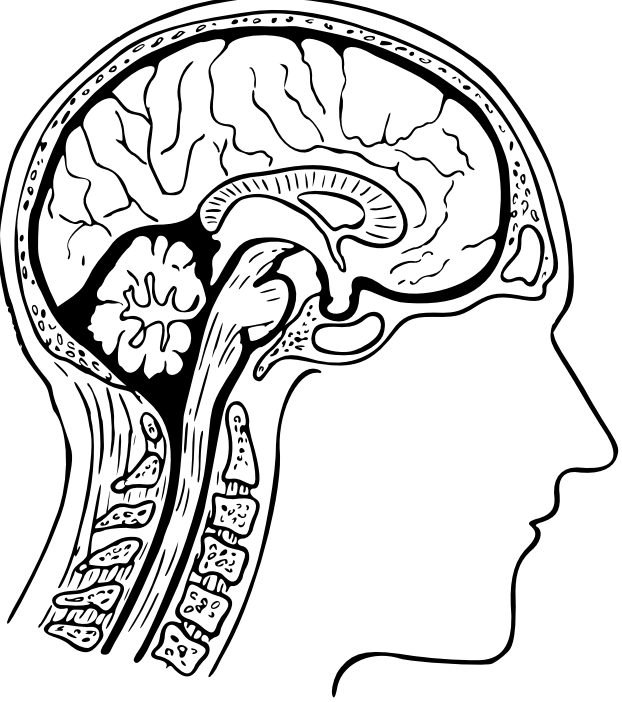
Part C: Live Coding in MNE



THE ARTIFACT REJECTION PIPELINE

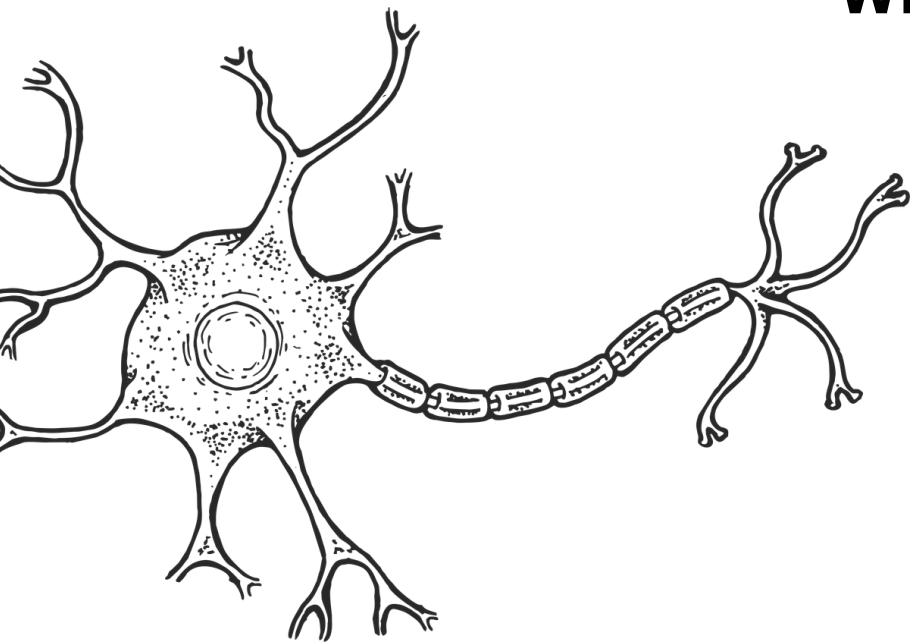
- Highpass Filter: Protect stationarity by applying a 1 Hz filter to a continuous copy.
- Fit ICA: Initialize and fit `mne.preprocessing.ICA` using the `fastica` algorithm.
- Identify Bad Components: Use `find_bads_eog()` to mathematically correlate components with the dedicated ocular channel.
- Reconstruct Signal: Zero out the specific bad components and apply the unmixing matrix to the Epochs object.



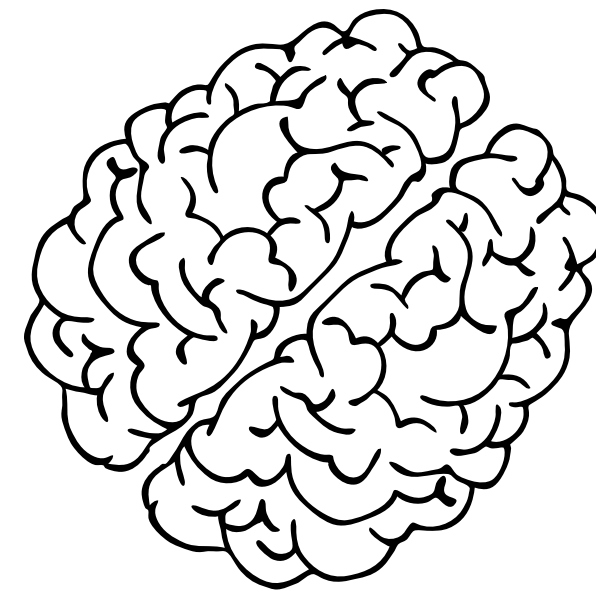


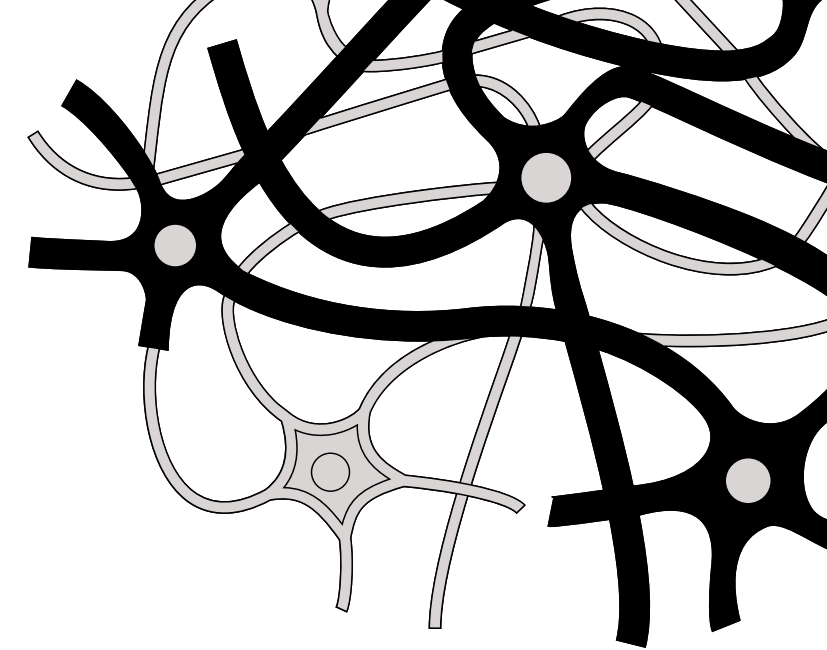
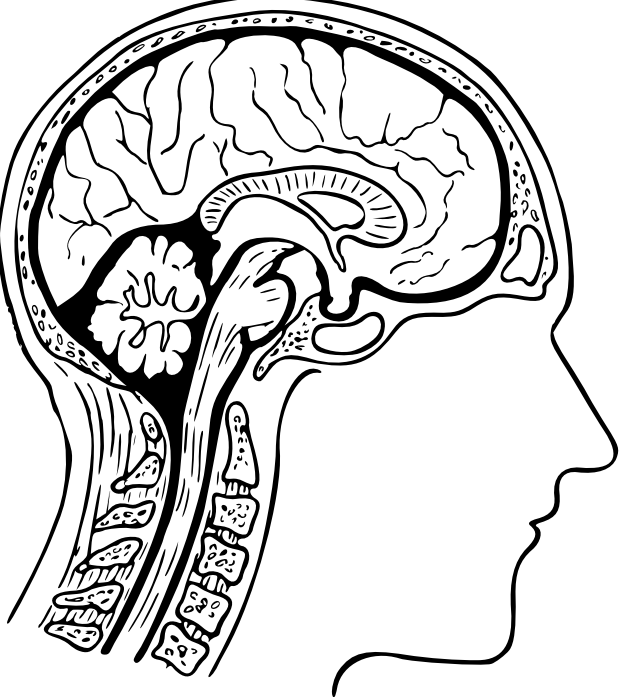
Evaluating the Reconstruction

**A successful ICA execution will remove the massive frontal deviation caused by blinks,
while leaving early visual Evoked Potentials completely intact.**



Colab





**Thank
You**

